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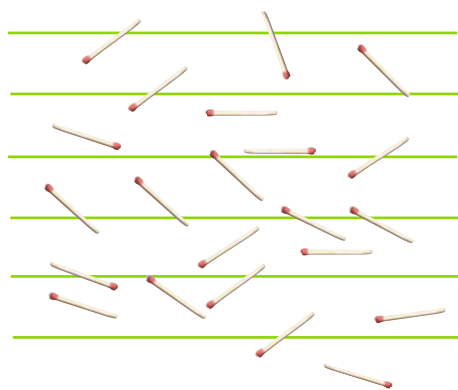
The best bet

Ever wondered why casinos make so much money? The answer is they make sure the odds are stacked against you. Look for the green zero on a roulette wheel. When the ball lands there, nobody wins. You have a 1 in 37 chance of winning on a number, but the casino only pays 36 times your bet. So on average, they always win.



Pi sticks

There's an interesting link between probability and pi (π). Drop matchsticks on a grid of lines one matchstick apart. The chance of a match touching a line is $2/\pi$, or about 0.64. Below, of 22 matches, around 14 should lie on a line (since $0.64 \times 22 = 14$). Try it for yourself.



TAKE A

What's the chance of being struck by lightning or hit by a meteorite when you go for a walk? If you fly in a plane, what's the chance of crashing or seeing a flying pig though a window? To answer these questions precisely, you need a branch of maths called probability.

WHAT IS PROBABILITY?

Probability is expressed by a number from zero to one. A probability of *zero* means something definitely won't happen, whereas a probability of *one* means it definitely will.

Anything in between means something *may* happen. For

instance, the chance of a coin landing heads up is a half, or 0.5.



MENDEL'S NUMBERS

In the 1850s, Austrian monk Gregor Mendel made an amazing discovery thanks to probability. Mendel bred purple-flowered peas with white-flowered peas and found that all the offspring were purple. He decided these must have "white" in them, but it just wasn't showing. So he bred the offspring with each other. Now there were four possibilities. The new parents could pass on one purple each, a purple and a white, a white and a purple, or two whites. If purple was present, it would show over white, and so on average, only a quarter of the plants would be white. They were. Mendel had discovered genes.

